E-Parada: Post-Defense Panel Revisions & Action Plan

Comprehensive Roadmap for Paper Revisions, Backend Engineering, & UI/UX Overhaul
College of Engineering and Architecture — National University (NU) Laguna
Proponents: Bustria, Christian Jake C. • Gamad, Leticia C. • Ovilla, Shan Jenis C. • Valiña, Resha Lorraine P.

🚨 Evaluation Respondent Pivot (3–10–10 Rule) | ✓ Two-Factor Vehicle KYC (OR/CR + License) | ✓ Provider Barangay Clearance | ✓ 1-Hour Downpayment & No-Show Clause | ✓ Color-Coded UI Overhaul | ✓ Mandatory On-Site Video Testing

PART 1: MANUSCRIPT & PAPER REVISIONS (By Chapter & Section)

1. Chapter 1: Specific Objectives Revision (The Respondent Pivot)

CHAPTER 1

The Problem: The paper previously stated that only IT developers would evaluate the system. The panel strictly corrected this: IT developers evaluate technical code and architectural quality, but **actual stakeholders must evaluate the operational application**.

> *"Ang IT developer, ite-test lang po niya is yung technical ng program... pero yung APPLICATION mismo, dapat USERS ang magte-test niyan! User acceptance / user evaluation!" — Thesis Panel*

Required Textual Revision for Specific Objective 3:

- **Objective 3a (Technical Software Quality):** To test and evaluate the technical software product quality of the E-Parada ecosystem through purposive testing by **IT Developers and Software Engineering Professionals** using the **ISO/IEC 25010:2011 Software Product Quality Model** in terms of *Functional Suitability*, *Performance Efficiency*, and *Reliability*.
- **Objective 3b (User Acceptance & Usability):** To evaluate the user acceptance, usability, and operational suitability of the mobile applications and web portal among key stakeholders in Calamba City, Laguna—specifically **Ten (10) Drivers/Motorists**, **Ten (10) Parking Space Providers**, and **Three (3) System Administrators**—measuring *Usability*, *User Error Protection*, *Learnability*, and *Operability*.

2. Chapter 1: Scope & Limitations Alignment

CHAPTER 1

- **Payment Processing Modality:** Correct any ambiguity regarding cashless payments. State explicitly in Scope that E-Parada implements a dual-modality settlement framework: (1) Cashless payments via **PayMongo API v1 integrating GCash** with automated webhook verification, and (2) On-site **Cash settlement** with mutual confirmation handshakes. In Limitations, clarify that cashless payment is fully supported but dependent on third-party gateway uptime and active data connectivity.
- **Exclusion of E-Bikes and Bicycles:** Explicitly delimit in Chapter 1 that the platform strictly caters to **LTO-registered motor vehicles** (Cars, Sedans, SUVs, Vans, Pickups, and Motorcycles). Bicycles, e-bikes, and light electric scooters are excluded due to the lack of government plate numbers, registration certificates (OR/CR), and licensing requirements.
- **Downpayment & No-Show Policy:** Document the operational policy requiring an initial 1-hour downpayment upon reservation to protect parking owners from no-show cancellations.

3. Chapter 3: Methodology & Sampling Design (The 3–10–10 Rule)

CHAPTER 3

Dual-Tier Evaluation Design:

Evaluation Tier	Target Evaluators	Evaluation Instrument	Core Characteristics
Tier 1: Technical Software Quality	**Purposively Selected IT Developers** (Software Engineers)	ISO/IEC 25010:2011 Evaluation Instrument & Automated Stress Harness	Functional Suitability, Performance Efficiency, Reliability, ACID Locking (`FOR UPDATE`).
Tier 2: User Acceptance & Usability	**23 Stakeholder Respondents:** • 10 Drivers / Motorists • 10 Parking Space Providers • 3 System Administrators	Usability & System Acceptance Questionnaire (5-Point Likert Scale)	Recognizability, Learnability, Operability, User Error Protection, UI Aesthetics.

Sampling Technique: Purposive Sampling based on verified driver licensing, parking property ownership/custody, and institutional administration credentials in Calamba City, Laguna.

PART 2: SYSTEM & SOFTWARE ENGINEERING REVISIONS

1. User Roles & Vehicle Ownership Differentiation

BACKEND / DATABASE

Problem Identified by Panel: The app assumed the driver is always the vehicle owner. What if someone drives a family car or company vehicle?

- **Registration Options:** Allow users to register as:
  1. **Driver Only:** Submits a valid **Driver's License**.
  2. **Vehicle Owner Only:** Submits Government ID + **OR/CR in their name**.
  3. **Owner & Driver (Both):** Submits both Driver's License and OR/CR.
  4. **Parking Space Provider:** Routes to property onboarding.

2. Two-Factor Vehicle & Space KYC Verification

SECURITY & VERIFICATION

> *"Paano ive-verify kung hindi nakaw? Baka nakapark doon nakaw pala, demandahan pa yung may-ari nung parking!... Hindi pwedeng Valid ID lang sa provider, laging dalawa ang verification!" — Thesis Panel*

- **Vehicle Registration KYC:** Drivers must upload:
  1. Vehicle photo with visible license plate (for **Tesseract OCR** plate extraction).
  2. Official **Certificate of Registration (CR)** issued by the LTO.
  3. Current-year **Official Receipt (OR)** issued by the LTO.
  4. Valid **Driver's License**.
- **Parking Space Provider KYC:** Providers must upload:
  1. Valid Government ID (with matching home address).
  2. **Barangay Clearance / Certificate of Parking Operation** issued by the local Calamba barangay explicitly authorizing parking operations at that specific address.
  3. *(Optional upload slot):* Real Property Tax Declaration or Land Title (TCT).
- **Admin Portal Review Interface:** Provide side-by-side verification tabs allowing administrators to inspect the vehicle photo, OCR plate, OR/CR document, and license before issuing approval.

3. 1-Hour Downpayment & 1-Hour No-Show Cancellation Logic

RESERVATION & BILLING

Problem: Drivers who book a slot and never show up tie up the space and cause revenue loss for the provider.

- **Mandatory 1-Hour Downpayment:** When reserving, the driver must immediately pay a downpayment equal to **1 hour of parking** (e.g., ₱50 via GCash PayMongo).
- **Automated 60-Minute No-Show Cancellation:** If the driver fails to arrive and scan the entry QR code within **60 minutes of the scheduled start time**:
  - The reservation is automatically cancelled (`Cancelled_NoShow`).
  - The slot is immediately returned to the available inventory.
  - The 1-hour downpayment is **forfeited by the driver and credited to the parking owner** as compensation.
- **Remaining Balance Settlement:** At exit, total duration is calculated: $ ext{Balance Due} = ext{Total Computed Bill} - ext{Downpayment Paid}$, payable via Cash or GCash.

4. Space Management: Batch Slot Configuration ("Select All")

PROVIDER PORTAL

Problem: Adding 15, 50, or 100 slots one by one is exhausting and impractical for providers.

- Implement a **"Select All" / Checkbox feature** in the parking slot editor.
- Allows providers to batch-assign vehicle types (e.g., set all to "Car/Sedan" or "SUV"), dimensions, and clearance with one click.

5. Communication Rules: Deactivate Chat on Inactive Bookings

WEBRTC & CHAT

- Hide and deactivate the in-app chat and WebRTC call buttons if the reservation is **Cancelled**, **Rejected**, or **Completed**. Communication is strictly reservation-bound.

6. Front-End Mobile & Web UI/UX Overhaul

MOBILE UI / UX

The panel strongly criticized the visual clutter, tiny text, and monochromatic color scheme in the demo:

UI / UX Dimension	Panel Observation	Required Technical Implementation
Functional Color-Coding	*"Puro blue, ang dumi tingnan... walang distinction."*	• **Green buttons:** Approve, Confirm, Pay Now. • **Red buttons:** Cancel, Reject, Delete, Overstay alert. • **Neutral slate/white:** Card containers and headers.
Typography & Readability	*"Napakaliit ng text... paano kung matanda ang driver o owner? Hahabulin ng magnifying glass?"*	• Enlarge base body text from 11sp to **14sp–16sp**. • Headers set to **18sp–22sp Bold**. • Minimum touch target height of **48dp** on all buttons.
Screen Clutter & Architecture	*"Ang gulo... ang daming nakasulat na hindi maintindihan kung ano ang pipindutin."*	• **Separate History Tab:** Move past/cancelled bookings to a dedicated History screen. • Main screen displays ONLY the currently active reservation card.
Card Presentation	Raw text dumped across the entire screen.	• **Cascading / Collapsible Cards:** Clean summary cards that smoothly expand when clicked to reveal details and 2D floor plans.

PART 3: CAPSTONE 3 COMPLIANCE & DEFENSE PROTOCOL

Mandatory Requirements for Capstone 3 Final Defense

CRITICAL MANDATE

> *"Kumuha kayo ng VIDEO, hindi lang picture! Kailangan videohan ninyo na ginagamit nila yung system sa actual parking space. Pagbalik niyo sa Capstone 3 at walang video, papaulitin ko kayo ng defense!" — Lead Panelist*

1. **On-Site Video Documentation:**
   - During the term break, visit actual parking lots/driveways in Calamba City.
   - Record high-definition video of real drivers and real lot owners conducting live reservations, QR code gate scans, and checkout billing.
   - Produce a 2–3 minute video presentation reel for the final defense.
2. **Elimination of Dummy Demo Data:**
   - Purge placeholder names (like "Rasha"), generic text, and sample images.
   - Populate the database with authentic Philippine vehicle models, realistic plate photos (`ABC-1234`), genuine Calamba street addresses, and real sample OR/CR and Driver's Licenses.
3. **Three-Device Live Demonstration Setup:**
   - **Device 1 (Web / Laptop):** System Administrator Portal (live KYC audits and telemetry).
   - **Device 2 (Android Phone A):** Parking Space Provider App (camera QR scanner and slot manager).
   - **Device 3 (Android Phone B):** Driver Mobile App (interactive map search, slot booking, and QR pass viewer).

PART 4: SUMMARY COMPLIANCE CHECKLIST

Item	Category	Specific Action Item	Target Scope
1	**Paper (Ch 1)**	Split Specific Objective 3 into Technical Software Quality (IT Devs) & User Usability (End-Users).	Manuscript
2	**Paper (Ch 1)**	Explicitly state Cash + Cashless support in Scope; remove cashless from limitations.	Manuscript
3	**Paper (Ch 1)**	Delimit vehicle scope: strictly LTO-registered motor vehicles; remove e-bikes and bicycles.	Manuscript
4	**Paper (Ch 3)**	Adopt the **3–10–10 Rule**: 10 Drivers, 10 Parking Providers, 3 Admins + IT Developer panel.	Manuscript
5	**Paper (Ch 3)**	Update DFD Level-1 to reflect OR/CR, Driver's License, Barangay Clearance, and Downpayment flow.	Manuscript
6	**System (Backend)**	Add OR/CR and Driver's License upload fields and side-by-side admin audit screen.	Laravel API / DB
7	**System (Backend)**	Add Barangay Clearance requirement for Parking Space Provider verification.	Laravel API / DB
8	**System (Backend)**	Implement 1-Hour Downpayment (GCash PayMongo) + 60-Minute No-Show auto-cancellation logic.	Billing Engine

Item	Category	Specific Action Item	Target Scope
9	**System (Backend)**	Purge e-bike / bicycle options from database seeds, validation rules, and UI dropdowns.	Database / API
10	**System (Mobile)**	Add "Select All" / batch slot configuration checkbox in provider space editor.	Flutter App
11	**System (Mobile)**	Disable/hide in-app chat and WebRTC call buttons for cancelled or rejected bookings.	Flutter App
12	**System (Mobile)**	Overhaul UI colors: Green for Approve/Pay, Red for Cancel/Reject, distinct header cards.	Flutter App
13	**System (Mobile)**	Enlarge typography (14sp–16sp body text) and touch targets (48dp minimum).	Flutter App
14	**System (Mobile)**	Move past bookings to a separate History tab; implement collapsible cascading cards.	Flutter App
15	**Capstone 3 Prep**	Record high-definition on-site video testing at actual Calamba City parking locations.	Field Testing
16	**Capstone 3 Prep**	Purge placeholder test data; populate with realistic Philippine IDs and OR/CR records.	Database Seed